Which mechanisms account for each dataset?

Eight mechanisms added to skill and practice, compared on correct RTs (12 percentiles), accuracy and error RTs in 16 contrasts; extended Wong-Wang model.

Best combination for each contrast adequate	14 of 16
Skill + practice alone adequate	4 of 16
One combination adequate in all 16	no
Shared combination adequate in the most contrasts	SP + evidence + threshold (WW + LCA) + lapse: 13 of 16
Shared combination with the lowest total AIC	SP + evidence + threshold (WW + LCA) + lapse (fails: P8;F1;S1)
Mechanism most often the best single addition	common drive 5, evidence 5, threshold (WW + LCA) 2

Per contrast

Id	Task	Change	Chosen combination	p	Best single
P1	brightness…	Time on task	common drive + evidence	0.91	common drive
P2	lexical decision	Time on task	E/I control	0.057	E/I control
P3	lexical decision (BLP)	Time on task	common drive	1	common drive
P4	motion discrimination	Time on task	threshold (WW + LCA)	1	threshold
P5	visual search	Time on task	(skill + practice only)	0.96	common drive
P6	visual working memory	Time on task	evidence	0.99	evidence
P7	Stroop and Simon	Time on task	common drive	0.83	common drive
P8	cued task switching	Time on task	evidence + urgency	0.49	evidence
P9	attention network…	Time on task	(skill + practice only)	0.93	urgency
A1	brightness…	Instruction	non-decision time + lapse + threshold (WW +…	0.94	non-decision time
A2	lexical decision	Instruction	threshold (WW + LCA) + lapse + evidence	0.65	threshold
F1	lexical decision	Word frequency	evidence + threshold (WW + LCA) + E/I…	0.022 (no)	evidence
S1	StarCraft 2 by league	Expertise	common drive	0.00013 (no)	common drive
S2	motion discrimination	Expertise	non-decision time	1	non-decision time
D1	motion discrimination…	Stimulus strength	evidence + non-decision time + common drive	1	evidence
D2	motion discrimination	Stimulus strength	evidence + threshold (WW + LCA)	0.93	evidence

1. Chosen combination: forward selection from skill + practice, adding the mechanism with the lowest AIC while AIC falls by more than 2. Adequate: chi-square p > .05; (no) = not adequate. Best single: the mechanism that lowers AIC most when added alone.

2. Mechanisms: common drive (u0, u0_sd scaled); evidence (pool difference); E/I control (strength, ratio; Lo, Wang & Wang 2015); threshold of the circuit and of the instance route; urgency (input growing through the trial); lapse (fraction of trials with reduced drive and no evidence); non-decision time; retrieval accuracy (instance-route bias).

3. Shared combinations: every combination chosen for any contrast, and their union, fitted to all 16 (Figure 5). P3 and S1 have no accuracy (12 terms), D1 has no error RTs in B (13 terms), so models with as many changes as terms have no p there.

4. Caution: each fit's chi-square includes its own simulation error. Where the data's accuracy error is very small (D1 reaches 100% correct), a model with noisier simulated accuracy is penalised less; this is why some single additions show a negative AIC drop in Figure 1 (D1, D2). Comparisons there are not reliable.

5. Sources: 4_output/mechanism_comparison.csv, shared_combinations.csv, mech_selection/, shared_combinations/.

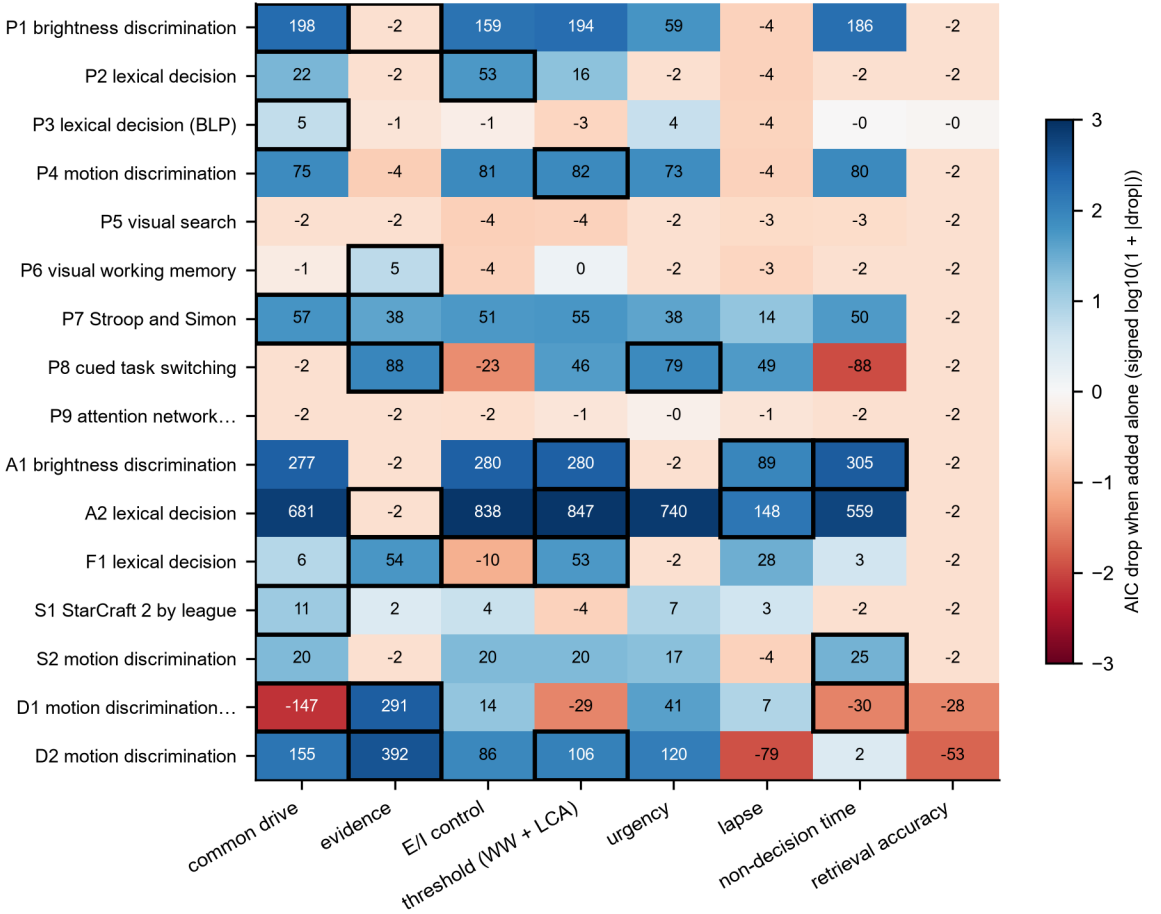


Figure 1. What each mechanism does on its own: the drop in AIC when it alone is added to skill + practice (numbers; blue = better fit). Black boxes: mechanisms in the combination chosen for that contrast. A negative drop (red) cannot be a worse fit of a larger model; it comes from each fit's own simulation error in the chi-square (summary, note 5).

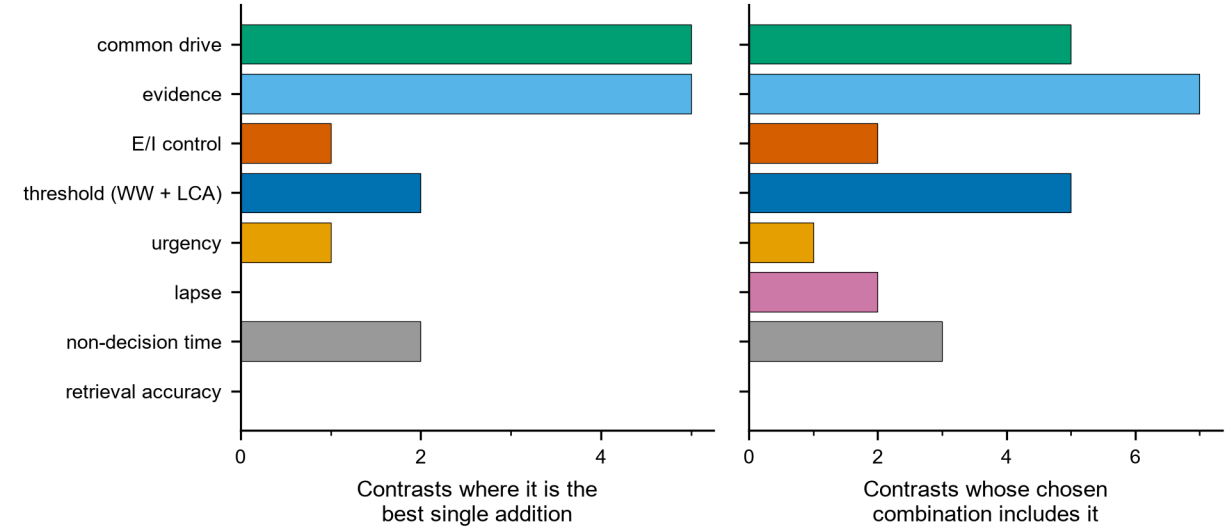


Figure 2. How often each mechanism is needed, out of 16 contrasts.

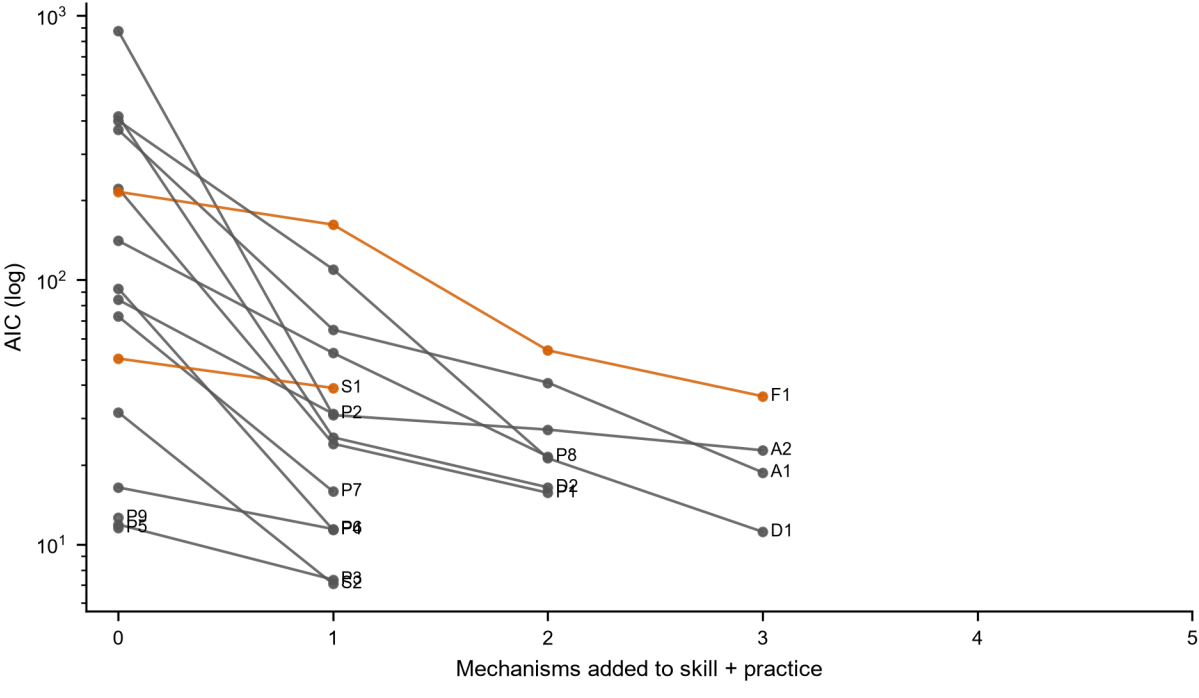


Figure 3. AIC along each contrast's selection path, one mechanism added per step until AIC no longer falls by more than 2. Grey: the chosen combination is adequate; orange: it is not.

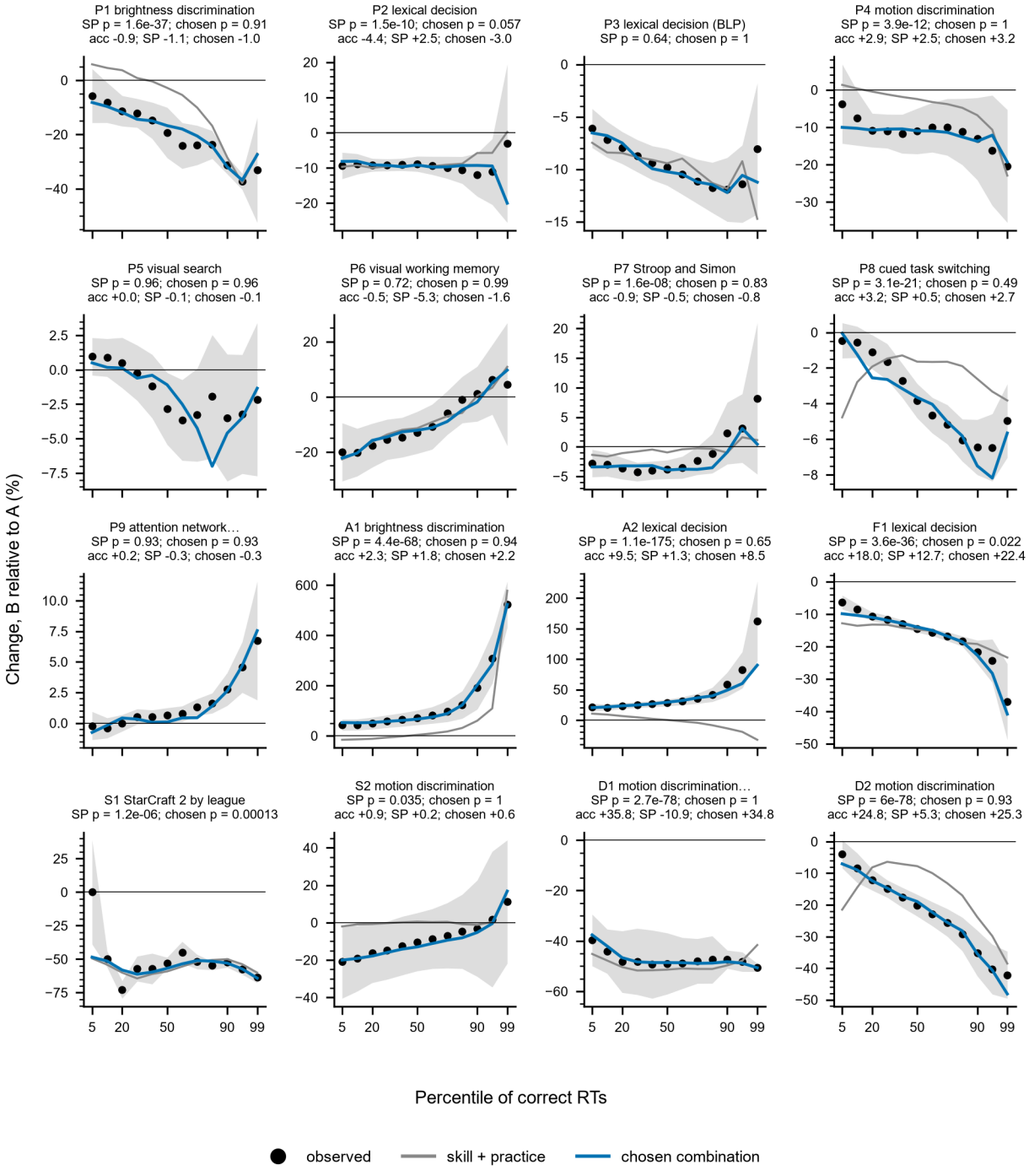


Figure 4. Observed change in 12 correct-RT percentiles (points; band = 95% interval), skill + practice (grey) and the chosen combination (blue). Titles: chi-square p and the observed and predicted change in accuracy (points).

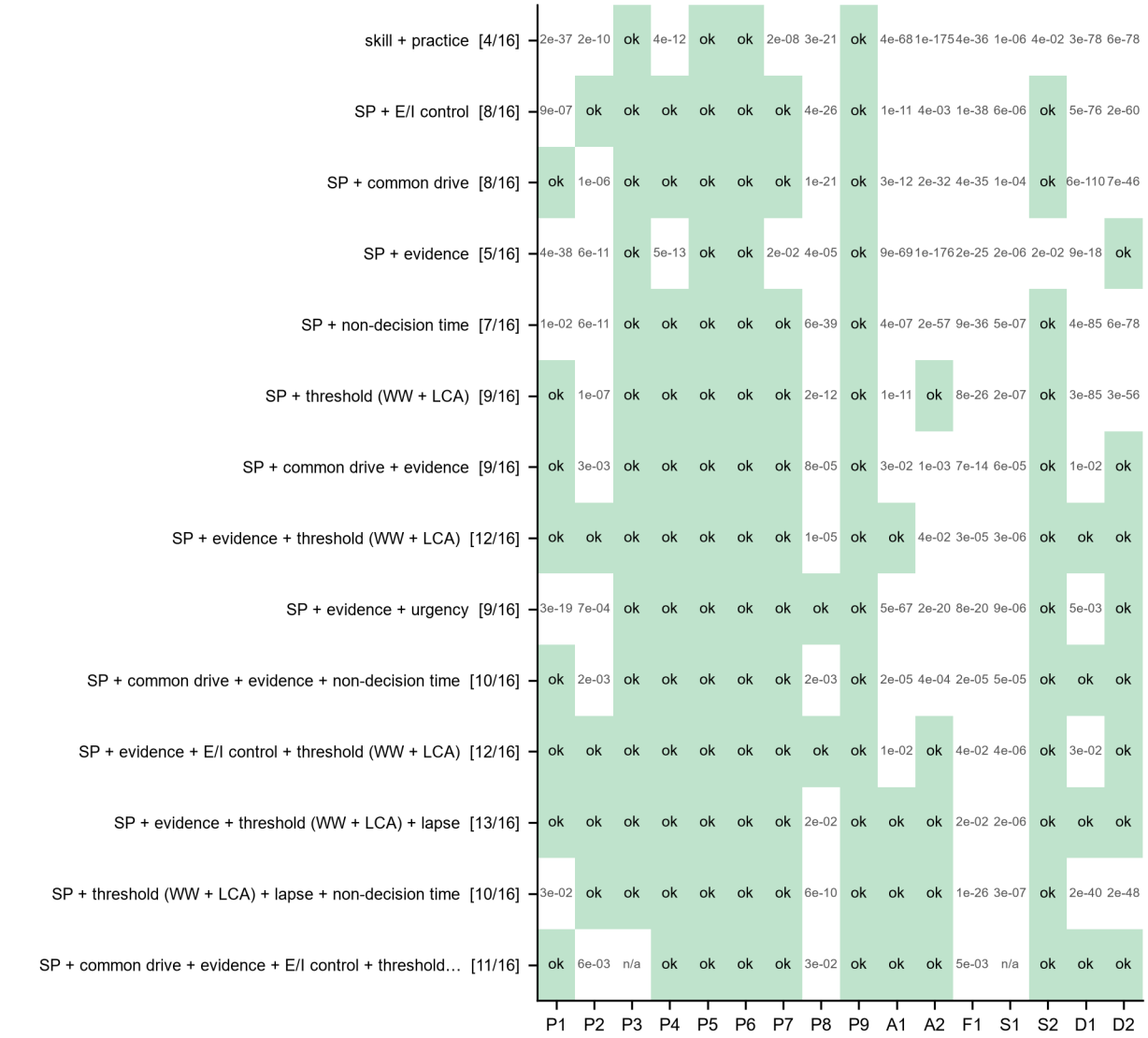


Figure 5. Each combination chosen for any contrast, and their union (last row), fitted to every contrast. Green 'ok' = adequate ($p > .05$); otherwise p, or n/a where the model has as many changes as observations. Brackets: contrasts fitted adequately.